

7. Conservation of Momentum

In this section the law of conservation of momentum will be expressed in the differential form directly by applying Newton's law of motion to an infinitesimal fluid element. We shall then show how the differential form could be derived by starting from an integral form of Newton's law.

Consider the motion of the infinitesimal fluid element shown in Figure 4.7. Newton's law requires that the net force on the element must equal mass times the acceleration of the element. The sum of the surface forces in the x_1 direction equals

$$\begin{aligned} & \left(\tau_{11} + \frac{\partial \tau_{11}}{\partial x_1} \frac{dx_1}{2} - \tau_{11} + \frac{\partial \tau_{11}}{\partial x_1} \frac{dx_1}{2} \right) dx_2 dx_3 \\ & + \left(\tau_{21} + \frac{\partial \tau_{21}}{\partial x_2} \frac{dx_2}{2} - \tau_{21} + \frac{\partial \tau_{21}}{\partial x_2} \frac{dx_2}{2} \right) dx_1 dx_3 \\ & + \left(\tau_{31} + \frac{\partial \tau_{31}}{\partial x_3} \frac{dx_3}{2} - \tau_{31} + \frac{\partial \tau_{31}}{\partial x_3} \frac{dx_3}{2} \right) dx_1 dx_2, \end{aligned}$$

which simplifies to

$$\left(\frac{\partial \tau_{11}}{\partial x_1} + \frac{\partial \tau_{21}}{\partial x_2} + \frac{\partial \tau_{31}}{\partial x_3} \right) dx_1 dx_2 dx_3 = \frac{\partial \tau_{j1}}{\partial x_j} dV,$$

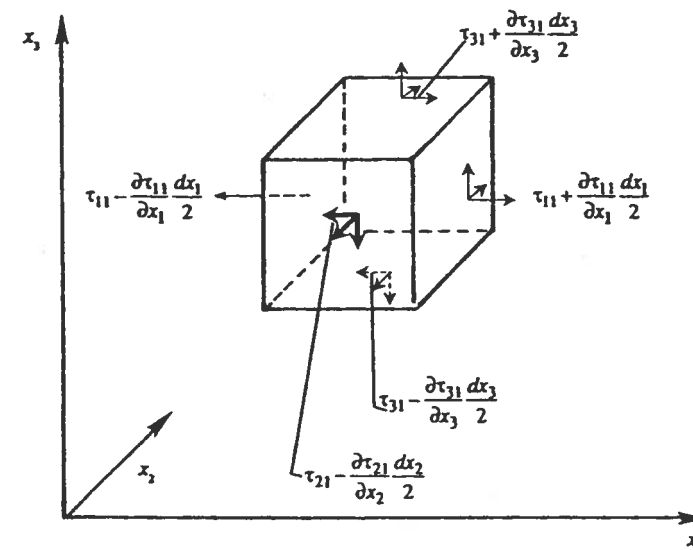


Figure 4.7 Surface stresses on an element moving with the flow. Only stresses in the x_1 direction are labeled.

where dV is the volume of the element. Generalizing, the i -component of the surface force per unit volume of the element is

$$\frac{\partial \tau_{ij}}{\partial x_j},$$

where we have used the symmetry property $\tau_{ij} = \tau_{ji}$. Let $\mathbf{g}$ be the body force per unit mass, so that $\rho \mathbf{g}$ is the body force per unit volume. Then Newton's law gives

$$\rho \frac{Du_i}{Dt} = \rho g_i + \frac{\partial \tau_{ij}}{\partial x_j}. \quad (4.15)$$

This is the equation of motion relating acceleration to the net force at a point and holds for any continuum, solid or fluid, no matter how the stress tensor τ_{ij} is related to the deformation field. Equation (4.15) is sometimes called *Cauchy's equation of motion*.

We shall now deduce Cauchy's equation starting from an *integral* statement of Newton's law for a material volume V . In this case we do not have to consider the internal stresses within the fluid, but only the surface forces at the boundary of the volume (along with body forces). It was shown in Chapter 2, Section 6 that the surface force per unit area is $\mathbf{n} \cdot \boldsymbol{\tau}$, where $\mathbf{n}$ is the unit outward normal. The surface force on an area element dA is therefore $d\mathbf{A} \cdot \boldsymbol{\tau}$. Newton's law for a material volume V requires that the rate of change of its momentum equals the sum of body forces throughout the volume, plus the surface forces at the boundary. Therefore

$$\frac{D}{Dt} \int_V \rho u_i dV = \int_V \rho \frac{Du_i}{Dt} dV = \int_V \rho g_i dV + \int_A \tau_{ij} dA_j, \quad (4.16)$$

where equations (4.6) and (4.14) have been used. Transforming the surface integral to a volume integral, equation (4.16) becomes

$$\int \left[\rho \frac{Du_i}{Dt} - \rho g_i - \frac{\partial \tau_{ij}}{\partial x_j} \right] dV = 0.$$

As this holds for any volume, the integrand must vanish at every point and therefore equation (4.15) must hold. We have therefore derived the differential form of the equation of motion, starting from an integral form.

8. Momentum Principle for a Fixed Volume

In the preceding section the momentum principle was applied to a *material* volume of finite size and this led to equation (4.16). In this section the form of the law will be derived for a fixed region in space. It is easy to do this by starting from the differential form (4.15) and integrating over a fixed volume V . Adding u_i times the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) = 0,$$

to the left-hand side of equation (4.15), we obtain

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) = \rho g_i + \frac{\partial \tau_{ij}}{\partial x_j}. \quad (4.17)$$

Each term of equation (4.17) is now integrated over a fixed region V . The time derivative term gives

$$\int_V \frac{\partial(\rho u_i)}{\partial t} dV = \frac{d}{dt} \int_V \rho u_i dV = \frac{dM_i}{dt}, \quad (4.18)$$

where

$$M_i \equiv \int_V \rho u_i dV,$$

is the momentum of the fluid inside the volume. The volume integral of the second term in equation (4.17) becomes, after applying Gauss' theorem,

$$\int_V \frac{\partial}{\partial x_j}(\rho u_i u_j) dV = \int_A \rho u_i u_j dA_j \equiv \dot{M}_i^{\text{out}}, \quad (4.19)$$

where $\dot{M}_i^{\text{out}}$ is the net rate of outflux of i -momentum. (Here $\rho u_j dA_j$ is the mass outflux through an area element dA on the boundary. Outflux of momentum is defined as the outflux of mass times the velocity.) The volume integral of the third term in equation (4.17) is simply

$$\int \rho g_i dV = F_{bi}, \quad (4.20)$$

where F_b is the net body force acting over the entire volume. The volume integral of the fourth term in equation (4.17) gives, after applying Gauss' theorem,

$$\int_V \frac{\partial \tau_{ij}}{\partial x_j} dV = \int_A \tau_{ij} dA_j \equiv F_{si}, \quad (4.21)$$

where F_s is the net surface force at the boundary of V . If we define $\mathbf{F} = \mathbf{F}_b + \mathbf{F}_s$ as the sum of all forces, then the volume integral of equation (4.17) finally gives

$$\mathbf{F} = \frac{d\mathbf{M}}{dt} + \dot{\mathbf{M}}^{\text{out}}, \quad (4.22)$$

where equations (4.18)–(4.21) have been used.

Equation (4.22) is the law of conservation of momentum for a fixed volume. It states that the net force on a fixed volume equals the rate of change of momentum within the volume, plus the net outflux of momentum through the surfaces. The equation has three independent components, where the x -component is

$$F_x = \frac{dM_x}{dt} + \dot{M}_x^{\text{out}}.$$

The momentum principle (frequently called the *momentum theorem*) has wide application, especially in engineering. An example is given in what follows. More illustrations can be found throughout the book, for example, in Chapter 9, Section 4, Chapter 10, Section 11, Chapter 13, Section 10, and Chapter 16, Sections 2 and 3.

Example 4.1. Consider an experiment in which the drag on a 2D body immersed in a steady incompressible flow can be determined from measurement of the velocity distributions far upstream and downstream of the body (Figure 4.8). Velocity far upstream is the uniform flow U_∞ , and that in the wake of the body is measured to be $u(y)$, which is less than U_∞ due to the drag of the body. Find the drag force D per unit length of the body.

Solution: The wake velocity $u(y)$ is less than U_∞ due to the drag forces exerted by the body on the fluid. To analyze the flow, take a fixed volume shown by the dashed lines in Figure 4.8. It consists of the rectangular region PQRS and has a hole in the center coinciding with the surface of the body. The sides PQ and SR are chosen far enough from the body so that the pressure nearly equals the undisturbed pressure p_∞ . The side QR at which the velocity profile is measured is also at a far enough distance for the streamlines to be nearly parallel; the pressure variation across the wake is therefore small, so that it is nearly equal to the undisturbed pressure p_∞ . The surface forces on PQRS therefore cancel out, and the only force acting at the boundary of the chosen fixed volume is D , the force exerted by the body at the central hole.

For steady flow, the x -component of the momentum principle (4.22) reduces to

$$D = \dot{M}^{\text{out}}, \quad (4.23)$$

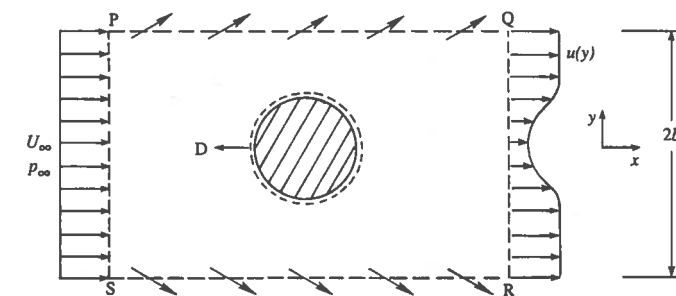


Figure 4.8 Momentum balance of flow over a body (Example 4.1).

where $\dot{M}^{\text{out}}$ is the net outflow rate of x -momentum through the boundaries of the region. There is no flow of momentum through the central hole in Figure 4.8. Outflow rates of x -momentum through PS and QR are

$$\dot{M}^{\text{PS}} = - \int_{-b}^b U_{\infty} (\rho U_{\infty} dy) = -2b\rho U_{\infty}^2, \quad (4.24)$$

$$\dot{M}^{\text{QR}} = \int_{-b}^b u (\rho u dy) = \rho \int_{-b}^b u^2 dy. \quad (4.25)$$

An important point is that there is an outflow of mass and x -momentum through PQ and SR. A mass flux through PQ and SR is required because the velocity across QR is less than that across PS. Conservation of mass requires that the inflow through PS, equal to $2b\rho U_{\infty}$, must balance the outflows through PQ, SR, and QR. This gives

$$2b\rho U_{\infty} = \dot{m}^{\text{PQ}} + \dot{m}^{\text{SR}} + \rho \int_{-b}^b u dy,$$

where $\dot{m}^{\text{PQ}}$ and $\dot{m}^{\text{SR}}$ are the outflow rates of mass through the sides. The mass balance can be written as

$$\dot{m}^{\text{PQ}} + \dot{m}^{\text{SR}} = \rho \int_{-b}^b (U_{\infty} - u) dy.$$

Outflow rate of x -momentum through PQ and SR is therefore

$$\dot{M}^{\text{PQ}} + \dot{M}^{\text{SR}} = \rho U_{\infty} \int_{-b}^b (U_{\infty} - u) dy, \quad (4.26)$$

because the x -directional velocity at these surfaces is nearly U_{∞} . Combining equations (4.22)–(4.26) gives a net outflow of x -momentum of:

$$\dot{M}^{\text{out}} = \dot{M}^{\text{PS}} + \dot{M}^{\text{QR}} + \dot{M}^{\text{PQ}} + \dot{M}^{\text{SR}} = -\rho \int_{-b}^b u(U_{\infty} - u) dy.$$

The momentum balance (4.23) now shows that the body exerts a force on the fluid in the negative x direction of magnitude

$$D = \rho \int_{-b}^b u(U_{\infty} - u) dy,$$

which can be evaluated from the measured velocity profile.

A more general way of obtaining the force on a body immersed in a flow is by using the Euler momentum integral, which we derive in what follows. We must assume that the flow is steady and body forces are absent. Then integrating (4.17) over a fixed volume gives

$$\int_V \nabla \cdot (\rho \mathbf{u} \mathbf{u} - \boldsymbol{\tau}) dV = \int_A (\rho \mathbf{u} \mathbf{u} - \boldsymbol{\tau}) \cdot d\mathbf{A}, \quad (4.27)$$

where A is the closed surface bounding V . This volume V contains *only* fluid particles. Imagine a body immersed in a flow and surround that body with a closed surface. We seek to calculate the force on the body by an integral over a possibly distant surface. In order to apply (4.27), A must bound a volume containing only fluid particles. This is accomplished by considering A to be composed of three parts (see Figure 4.9),

$$A = A_1 + A_2 + A_3.$$

Here A_1 is the outer surface, A_2 is wrapped around the body like a tight-fitting rubber glove with $d\mathbf{A}_2$ pointing outwards from the fluid volume and, therefore, into the body, and A_3 is the connection surface between the outer A_1 and the inner A_2 . Now

$$\int_{A_3} (\rho \mathbf{u} \mathbf{u} - \boldsymbol{\tau}) \cdot d\mathbf{A}_3 \rightarrow 0 \quad \text{as } A_3 \rightarrow 0,$$

because it may be taken as the bounding surface of an evanescent thread. On the surface of a solid body, $\mathbf{u} \cdot d\mathbf{A}_2 = 0$ because no mass enters or leaves the surface. Here $\int_{A_2} \boldsymbol{\tau} \cdot d\mathbf{A}_2$ is the force the body exerts on the fluid from our definition of $\boldsymbol{\tau}$. Then the force the fluid exerts on the body is

$$\mathbf{F}_B = - \int_{A_2} \boldsymbol{\tau} \cdot d\mathbf{A}_2 = - \int_{A_1} (\rho \mathbf{u} \mathbf{u} - \boldsymbol{\tau}) \cdot d\mathbf{A}_1. \quad (4.28)$$

Using similar arguments, mass conservation can be written in the form

$$\int_{A_1} \rho \mathbf{u} \cdot d\mathbf{A}_1 = 0. \quad (4.29)$$

Equations (4.28) and (4.29) can be used to solve Example 4.1. Of course, the same final result is obtained when $\boldsymbol{\tau} \approx$ constant pressure on all of A_1 , $\rho =$ constant, and the x component of $\mathbf{u} = U_{\infty} \mathbf{i}$ on segments PQ and SR of A_1 .

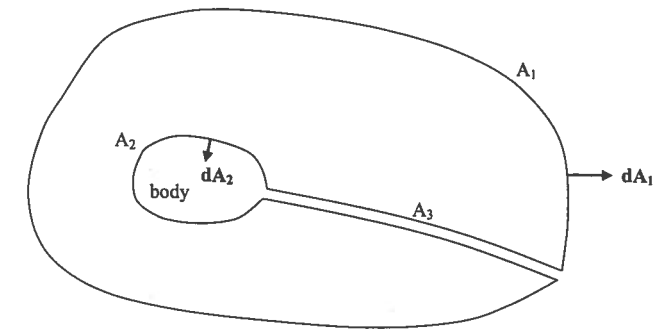


Figure 4.9 Surfaces of integration for the Euler momentum integral.